

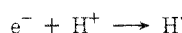
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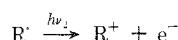
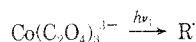
Introduction

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Co(II) and Co(III) complexes. Several authors have studied the photolysis of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ in frozen samples in order to accumulate the reactive intermediate species. Shagisultanova and Posnjak^{10,11} detected two radical species in their epr studies after uv irradiation of a solid $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ matrix at 77°K. One of the signals was identified as belonging to the $\text{H}\cdot$ radical. The other belongs to an unknown radical. In the beginning of the photolysis the unknown radical is predominantly formed. Later its signal decreases as the $\text{H}\cdot$ radical signal increases. Their experiments indicate that the $\text{H}\cdot$ radicals are formed by the reaction



The electrons are formed by the photoreduction of a radical species, formed subsequent to the photoreduction of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$



They conclude that $\text{R}\cdot$ is most probably not a free $\text{C}_2\text{O}_4^{\cdot-}$ species but rather an oxalate radical bound to the $\text{Co}(\text{C}_2\text{O}_4)_2^{2-}$ complex, similar to the radical proposed by Parker and Hatchard.⁴ The occurrence of these two radical species $\text{H}\cdot$ and $\text{R}\cdot$ was confirmed by the experiments of Eaton and Suart.¹²

Experimental Section

(a) *Apparatus.* A flash-photolysis apparatus of the conventional type, inspired by the one developed by Parker and Hatchard,⁴ has been constructed in the laboratory. As actinic light source two hydrogen-filled flash tubes were used. Two capacitors (20 μF each) could be charged to 10 kV. The flash duration was between 10 and 20 μsec . The reaction vessel was a fused quartz tube, length 20 cm and i.d. 1 cm, with optically flat windows sealed at each end.

Transmission was detected as a function of time, at constant wavelength. From these data absorbance values were calculated. As a detection light source a 450-W xenon lamp was used, fed by a current stabilized power supply. The detection system consisted of a Bausch & Lomb high-intensity monochromator and a RCA 1P28 photomultiplier tube. To improve the signal-to-noise ratio, only the first four dynodes were used. The others were electrically connected to the anode. The signals passed through an emitter follower (bandwidth 1 Mhz) and were recorded by means of an oscilloscope.

The experimental procedure followed is very similar to the one described by Parker and Hatchard.⁴

(b) *Preparation of the Solutions.* $\text{K}_3[\text{Co}(\text{C}_2\text{O}_4)_3] \cdot 3.5\text{H}_2\text{O}$ was prepared as described in the literature.¹³ The molar extinction coefficients found were $\epsilon_{600\text{ nm}}$ 162 and $\epsilon_{420\text{ nm}}$ 219, in good agreement with the literature.^{8,14} A cobalt analysis by atomic absorption spectrophotometry was likewise in agreement with the expected values.

Solutions were prepared by dissolving the 10^{-5} to 10^{-4} M $\text{K}_3[\text{Co}(\text{C}_2\text{O}_4)_3] \cdot 3.5\text{H}_2\text{O}$ in 3×10^{-3} M solutions of oxalic acid. The solutions were deaerated by bubbling with oxygen-free argon. All these manipulations were done with dimmed red light illumination.

Experimental Observations

In the experimental part as well as in the discussion two sets of data are to be clearly distinguished.

In the first set of experiments the initial concentration

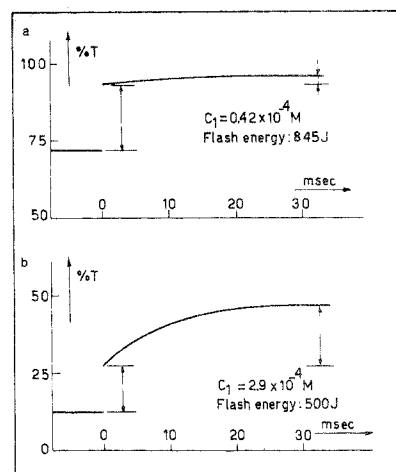


Figure 1. Transients at 600 nm for complete (a) and incomplete (b) immediate conversion of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$.

and the flash energy were chosen so as to convert with one flash discharge all the initially present $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ into intermediate species, thus ruling out all reactions of the intermediate species with $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$. Such conditions will be referred to as "experiments with complete immediate conversion." In our experimental set up the lowest initial concentrations used (4×10^{-5} M) required a minimal flash energy of 320 J; for a concentration of 1.1×10^{-4} M 845 J was necessary to have complete immediate conversion.

In a subsequent series of experiments the kinetics were investigated when an important amount of initial reagent remained unreacted immediately after the flash, allowing the intermediate species to react with this excess $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$. Such conditions will be referred to as "experiments with incomplete immediate conversion."

(1) *Experiments with Complete Immediate Conversion.* The spectrum of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ shows three absorption bands with maxima at 254 (ϵ 21,400), 420 (ϵ 218), and 600 nm (ϵ 165). It was observed that the intermediate species produced by the flash pulse do not absorb at 600 nm. Indeed, when, e.g., a 4.2×10^{-5} M solution is photolyzed with a flash of 845 J, the absorbance at 600 nm drops to zero within the time duration of the flash, and remains negligible throughout the longest observation times (see Figure 1a). This means that all the $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ is immediately converted into intermediate species. The absorbance observed in the 320–550-nm region is, in this case, due only to the absorption of the intermediate species produced by the decomposition of the initial reagent.

Curve b of Figure 2 shows the spectrum observed at the end of the light burst when a 4.2×10^{-5} M $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ solution is flashed with an energy of 845 J. The intersection around 390 nm is no isosbestic point; when either the energy of the flash is increased, or the initial concentration of the solution is decreased, within the limits of complete immediate conversion, the intersection point shifts. This means that the transient absorption signal is due to at least two intermediate species, whose concentration ratio depends on the initial concentration ratio of the solution and on the flash energy.

The decay of the transient signals observed in the wavelength region from 320 to 600 nm has been measured in order to establish their kinetic parameters.

(2) *Experiments with Incomplete Immediate Conversion.* In these experiments the transient absorbance sig-

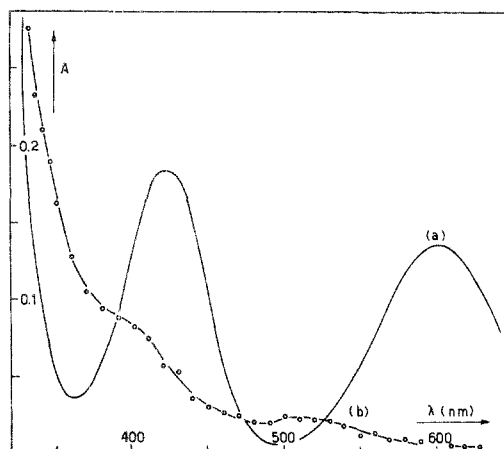


Figure 2. Absorption spectrum of a $4.2 \times 10^{-5} M$ $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ solution (a) before and (b) immediately after exposure to the flash (optical pathlength 20 cm).

nals result from the intermediate species as well as from the $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ remaining immediately after the action of the flash.

Again, 600 nm was used as a specific monitoring wavelength for $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$. Figure 2 showed indeed that the intermediate species do not interfere at this wavelength. As can be seen from the comparison of Figure 1b and Figure 1a, the shape of the transmission signal changes drastically when the primary photodecomposition is not complete. Subsequent to the sudden drop in the $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concentration due to the decomposition by the flash, an appreciable amount is consumed in secondary reactions.

In order to investigate the kinetics of these secondary reactions, the absorbance transients have been recorded for initial $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concentrations varying from 1 to $3 \times 10^{-4} M$, using flash energies from 320 to 720 J.

The wavelength region from 320 to 600 nm was investigated. The transients at 600 nm were not only used to study the disappearance of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ but also to calculate the spectral changes due to the intermediate species in the region where their spectra overlap with the one of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$. The absorbance due to $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ was subtracted from the measured signal using the 600-nm data and the absorptivity of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ at the wavelength under investigation.

Discussion of the Results

(A) *Complete Immediate Conversion of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$.* (1) *Decay of the Intermediate Species.* The kinetic interpretation of the data of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ photolysis is facilitated by the fact that the final reaction products $\text{Co}(\text{II})$ and CO_2 do not absorb at the wavelengths under investigation. When all the $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ is converted immediately during the flash, the transient spectrum can only be due to the intermediate reaction products. In conditions of complete immediate conversion, the decay of the transient absorbance signal at any chosen wavelength does not correspond to a simple first-order reaction. This is illustrated at 340 nm in Figure 3. The plot can be analyzed as composed of two first-order decays

$$A_t = A_{0,X} e^{-k_X t} + A_{0,Y} e^{-k_Y t}$$

where $A_{0,X}$ and $A_{0,Y}$ are the absorbance values of the intermediate species X and Y present at the end of the flash pulse ($t = 0$). This relation yields constant values for k_X and k_Y as can be seen in Table I, where some of the data

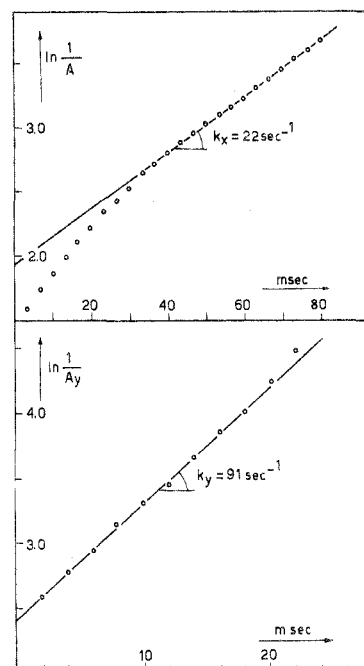


Figure 3. Rate constants for the decay of the intermediate species X (upper plot) and Y (lower plot).

TABLE I: Calculation of A_0 and k Values for the Intermediates X and Y in the Flash Photolysis of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$

Initial $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concn, M	Monitor- ing wave- length, nm	$A_{0,X}$	k_X , sec^{-1}	$A_{0,Y}$	k_Y , sec^{-1}
0.86×10^{-4}	330	0.230	20.0	0.238	85.3
	340	0.187	21.3	0.159	93.4
	350	0.127	20.0	0.144	90.0
	360	0.095	19.7	0.130	93.1
	370	0.070	20.0	0.135	99.1
0.71×10^{-4}	330	0.211	20.8	0.190	85.1
	340	0.148	18.6	0.152	75.2
	350	0.122	19.4	0.109	84.4
	360	0.081	20.0	0.089	82.5
	370	0.077	18.3	0.111	89.6
0.57×10^{-4}	330	0.173	20.0	0.145	73.4
	340	0.143	21.6	0.091	91.1
	350	0.103	18.3	0.079	86.0
	360	0.072	18.7	0.072	86.4
	370	0.072	18.7	0.072	86.4
0.42×10^{-4}	330	0.171	23.5	0.080	94.8
	340	0.117	23.5	0.064	77.7
	350	0.090	20.4	0.051	87.0

^a Flash energy = 845 J.

have been collected. The k_X and k_Y values thus obtained are independent of the monitoring wavelength and independent of the initial $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concentration. The average values obtained from some 40 experiments at different wavelengths between 320 and 600 nm and different initial concentrations of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ are $k_X = 20 \text{ sec}^{-1}$ and $k_Y = 87 \text{ sec}^{-1}$. Standard deviations of the mean are $\sigma_X = 1.6$ and $\sigma_Y = 5.4$. This means that two species are formed upon photolysis of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$, disappearing in two independent first-order reactions. It also shows that there are no other stable or reactive species formed that absorb between 320 and 600 nm.

Between 420 and 600 nm, $A_{0,X}$ becomes so small that it can be neglected as compared to $A_{0,Y}$. Consequently, the

TABLE II: Calculated Absorptivities and Initial Concentrations of the Intermediate Species X and Y for a Flash Energy of 845 J

$c_i \times 10^5, M$	$c_{0,X} \times 10^5, M$	$c_{0,Y} \times 10^5, M$	335 nm		340 nm		345 nm		355 nm	
			ϵ_X	ϵ_Y	ϵ_X	ϵ_Y	ϵ_X	ϵ_Y	ϵ_X	ϵ_Y
4.19	2.86	1.49	318	291	237	213	203	187	145	152
5.68	3.18	2.47	298	272	225	210			151	155
7.12	3.42	3.58	325	273			204	187	150	149
8.57	3.88	4.50	310	272	237	211			144	158
		Av values	313	277	233	211	204	187	148	154

$\ln 1/A$ vs. time plots give straight lines in the 420–600-nm region with a slope corresponding to k_Y . Previous experiments never outlined the presence of the two distinct intermediate species X and Y observed here. Intermediate X has probably not been detected in previous work since little attention has been paid to experimental conditions when all $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ is suddenly and completely converted into intermediate species. As will be shown below the ratio $A_{0,X}/A_{0,Y}$ decreases sharply when the initial $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concentration is increased.

Parker and Hatchard⁴ observed that their calculated first-order decay constants in the flash photodecomposition of $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ varied by a factor of 1.8 when the initial concentration varied by a factor of 20. This variation could well be due to the simultaneous decay of two species instead of the one proposed by these authors.

(2) *Absorptivities and Concentrations of the Intermediate Species.* When complete immediate conversion into intermediate species is achieved for a given flash energy and for a given initial concentration of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$, such complete conversion will of course also be obtained when higher flash energies are used. A systematic investigation revealed however that, although the conversion is complete for such experiments the absorbance immediately after the flash is found to be a function of the flash energy. In fact the ratio $A_{0,X}/A_{0,Y}$ depends on the flash energy as well as on the initial reagent concentration. High light intensity or low initial $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concentrations favor the formation of species X at the expense of species Y. Although it is in general not possible to calculate the concentrations of two simultaneously present intermediate species from absorbance data obtained in flash work, this concentration dependence turns the present system into one of the fortunate cases in which this can actually be done.

For two experiments with identical flash energy but different initial concentrations $c_{1,1}$ and $c_{1,2}$ one has, provided the immediate conversion is complete

$$c_{1,1} = (c_{0,X})_1 + (c_{0,Y})_1 \quad (3)$$

where $(c_{0,X})_1$ and $(c_{0,Y})_1$ are the concentrations of the intermediate species X and Y at time $t = 0$ and the initial concentration $c_{1,1}$ considered.

For the concentration $c_{1,2}$ one can write

$$c_{1,2} = a(c_{0,X})_1 + b(c_{0,Y})_1 \quad (4)$$

where the factors a and b have different values, since the concentration dependence of the two intermediates species on c_1 is different. If one of the intermediates had a binuclear structure the equations would have to be adapted in the proper way.

To calculate a and b the experimental values of $A_{0,X}$ and $A_{0,Y}$ are used. Indeed, one can write that at the concentrations $c_{1,1}$ and $c_{1,2}$

$$(A_{0,X})_1 = \epsilon_{X,\lambda}(c_{0,X})_1 l \quad (5)$$

$$(A_{0,X})_2 = \epsilon_{X,\lambda}a(c_{0,X})_1 l \quad (6)$$

where l is the pathlength of the cell, and $\epsilon_{X,\lambda}$ the extinction coefficient of species X at wavelength λ . $(A_{0,X})_1$ and $(A_{0,X})_2$ are the $A_{0,X}$ values corresponding to the initial concentrations $c_{1,1}$ and $c_{1,2}$. Dividing these, one obtains

$$a = (A_{0,X})_2/(A_{0,X})_1 \quad (7)$$

Similarly from the absorbance data for the species Y it follows that

$$b = (A_{0,Y})_2/(A_{0,Y})_1 \quad (8)$$

The concentration $c_{0,X}$ and $c_{0,Y}$ of the two intermediates for a given initial concentration of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ can thus be computed from expressions 3 and 4. From the corresponding A_0 values at different wavelengths $\epsilon_{X,\lambda}$ and $\epsilon_{Y,\lambda}$ are easily calculated. This procedure can be repeated for any set of absorbance values and should yield constant values for ϵ_X and ϵ_Y , if the assumptions made are correct. Such a set of calculations is collected in Table II, for a series of experiments with a flash energy of 845 J. Table II gives only part of the data. It has been extended to calculate the absorption spectra of the intermediate species. Figure 4 gives these spectra together with that of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$.

The formation of species X is inhibited by the presence of large initial concentrations of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$. High flash energies on the other hand, favor its formation. The dependence of the ratio $c_{0,Y}/c_{0,X}$ on the amount of initial reagent can be visualized in Figure 5, using the data of Table II. The following linear relationship is found: $c_{0,Y}/c_{0,X} = 1.39 \times 10^4 c_1$. This equation suggests that for high $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concentrations ($10^{-3} M$ or higher) the concentrations of X will be negligible as compared to Y.

(B) *Incomplete Immediate Conversion of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$.* As already mentioned, $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concentration vs. time plots change considerably whenever, instead of having complete immediate conversion, a large excess of reagent is left immediately after the flash. This is shown in Figure 1b. In this case there is an important secondary consumption of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ at a rate comparable to that of the disappearance of the intermediate species.

In order to find out whether this secondary reaction of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ is one in which one of the two intermediate species X or Y is involved, the kinetics of these intermediates were investigated in conditions of incomplete immediate conversion.

(1) *Mass Balance of the System.* When, e.g., a $2.9 \times 10^{-4} M$ solution is photolyzed with a 845-J flash, one obtains the following results: initial $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concentration $2.9 \times 10^{-4} M$, amount decomposed by the flash $1.55 \times 10^{-4} M$, initial concentration of Y $1.37 \times 10^{-4} M$, $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ decomposed by secondary reaction $1.07 \times 10^{-4} M$, $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ remaining after the experiment $0.29 \times 10^{-4} M$. The concentration of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ was calculated from the absorbance data at 600 nm. The concentra-

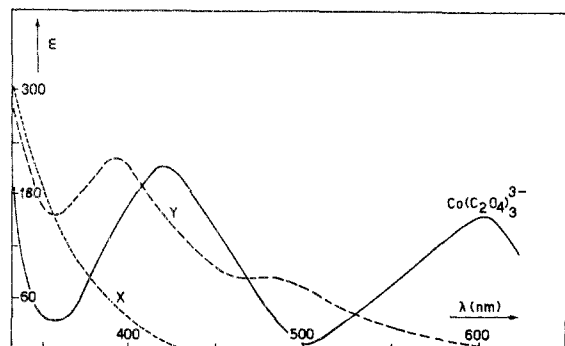


Figure 4. Spectrum of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$, and of the two intermediate species.

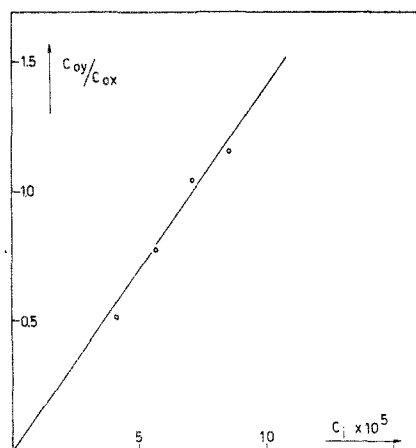


Figure 5. Ratio $C_{0,Y}/C_{0,X}$ as a function of the initial $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concentration for flash energy of 845 J.

tion of Y was obtained from the absorbance at 500 nm. Figure 4 shows indeed that at this wavelength Y is the only absorbing species.

It is clear that the concentration of X is low as compared to Y. Indeed, 88% of the $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ decomposed by the flash is converted into Y. This sets an upper limit of 12% to the concentration of X and all eventual other intermediates formed due to the presence of excess $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$.

(2) *Kinetic Behavior of Species X and Y in the Presence of Excess $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$.* The influence of an excess $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ on the kinetics of Y was investigated by measurements at 500 nm. The decay at Y under these circumstances remains strictly monomolecular with a rate constant of 80 sec^{-1} . The good fit with the k_Y value of 87 sec^{-1} , found from the fully independent complete conversion experiments, has to be emphasized. This proves indeed that the main pathway for the disappearance of Y is the same monomolecular reaction whether an excess $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ is present or not. This conclusion is in line with Parker and Hatchard's observation that no bimolecular process should be observed in the flash photolysis of $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$.⁴

It is however in obvious disagreement with the observation of Jamieson and Perone,⁷ who did not observe a monomolecular decay of the main intermediate species in the photolysis of $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$. The investigation of the kinetic behavior of X raises a problem. We have shown that for experiments with incomplete immediate conversion of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$, X is present in rather low concentrations (12%). The combination of these low concentrations

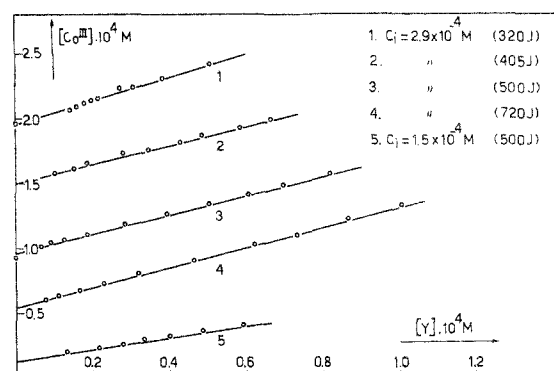
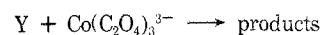


Figure 6. Concentration-concentration plots for $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ and Y at various initial $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concentrations and flash energies.

with the absorptivities of X, which are comparable to those of Y and $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$, simultaneously present in much larger concentrations, rules out the possibility for a direct kinetic study of X.

In the subsequent analysis of the system it is presumed that the decay rate constant for X in conditions of incomplete immediate conversion is identical with the one found for complete immediate conversion. It will be shown that this assumption is correct.

(3) *Kinetics of the Secondary $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ Consumption.* For the reason just mentioned a bimolecular reaction such as



is very improbable.

Nevertheless, there exists a relation between the rate of disappearance of Y and the rate of secondary $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ consumption. This relationship is shown in Figure 6, where for a number of initial $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concentrations and different flash energies the actual $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concentration is related to the corresponding concentration of the Y species in the same solution at various times after the flash. A linear relationship is found.

Its average slope is 0.9 except for the two experiments under conditions of near-complete immediate conversion. This means that for each species Y which disappears, $0.9 \text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ molecules are instantly reduced, independently of the eventual excess of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$.

The reaction rate for the secondary conversion of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ can thus be written as

$$d[\text{Co}(\text{C}_2\text{O}_4)_3^{3-}]/dt = 0.9 d[\text{Y}]/dt = 0.9 k_Y [\text{Y}]$$

To demonstrate this in a different way, the change in absorbance at 600 nm (where only $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ absorbs) due to this secondary conversion can be expressed as presented in Figure 7. A_t is the absorbance at time t and A_e is the final absorbance at $t = \infty$. $\ln 1/(A_t - A_e)$ is a linear function of t , with a slope $q_{\text{COIII}} = 75 \text{ sec}^{-1}$. The absorbance due to $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ at time t can be calculated from the equation

$$A_t = A_e + (A_0 - A_e) e^{-q_{\text{COIII}} t}$$

A_0 being the absorbance of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ immediately after the flash. The kinetics of the observed secondary conversion of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ can only be explained by a scheme in which the intermediary species Y decomposes into an unobserved species Z. This would in turn react very quickly and almost exclusively with $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$

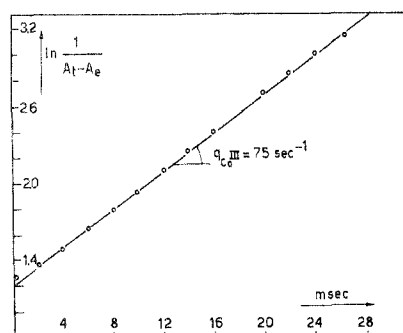
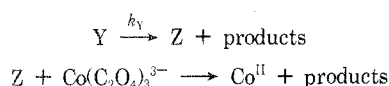


Figure 7. Calculation of the apparent rate constant for the reaction in which a secondary amount of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ is converted.



The last reaction must be very fast compared to the first. In this assumption Z should not be detectable spectroscopically since it should not accumulate. Even in experiments where a secondary conversion of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ is not possible, Z has not been detected. Two explanations are available for this: either the recombination reaction $Z + Z \rightarrow \text{products}$ is also fast, or, more probably, Z does not absorb in the investigated spectral region.

(4) *Recalculation of the Transient Signal at Any Chosen Wavelength.* If the conclusions resulting from the kinetic analysis are correct, it should be possible to recalculate any decay at any chosen wavelength using the absorptivities of the species $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$, X , and Y , the rate constants k_X and k_Y , the constant q_{CoIII} , and the concentrations of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$, X , and Y immediately after flash. Such a test is most elucidative when applied to incomplete immediate conversion experiments, since in this case one has to account for the secondary reduction of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ as well as for the disappearance of possible intermediate species.

The absorbance A_t at a time t after the flash would then be

$$A_t = A_e + A_{0,X} e^{-k_X t} + A_{0,Y} e^{-k_Y t} + (A_{0,\text{CoIII}} - A_e) e^{-q_{\text{CoIII}} t}$$

All A terms refer to absorbances at the wavelength λ . The various symbols have the following significance: A_t is the overall absorbance at any time t , $t = 0$ at the end of the light emission by the flash; A_e is the final absorbance due to the remaining $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ at the longest observation times; $A_{0,\text{CoIII}}$, $A_{0,X}$, and $A_{0,Y}$ are absorbances at time $t = 0$ due to respectively excess $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ and the intermediates X and Y ; q_{CoIII} , k_X , and k_Y are constants for the decay rates of the species referred to, the average values used are respectively 75, 20, and 87 sec^{-1} .

Figure 8 shows both the calculated and observed A_t values for a monitor wavelength of 340 nm, an initial $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concentration of $2.9 \times 10^{-4} M$ and 845-J dissipated flash energy. The good fit between the two kinds of values speaks favorably of the interpretation brought forward. It does even more so, if one keeps in mind that the values of k_X , k_Y , ϵ_X , and ϵ_Y used are the averages from the complete immediate conversion experiments.

(5) *Mechanism for the Photodecomposition of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$.* It is not possible to fit all these kinetic data into one of the mechanisms quoted in the Introduction. One should indeed account for the following observations.

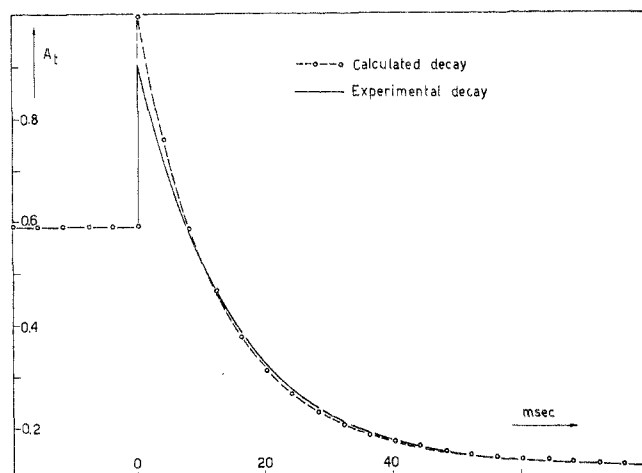
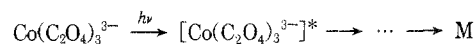


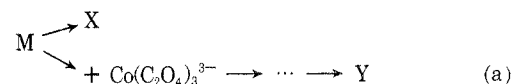
Figure 8. Comparison between a measured and a calculated transient signal at 340 nm.

(1) The $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ decomposed during the flash is transformed into two reactive intermediate species, X and Y . Their relative concentrations depend on both the concentration of the initial cobalt complex and the flash energy used. In conditions of complete immediate conversion both species are present in comparable concentrations. When excess $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ is available Y is the predominant form. (2) Both X and Y decay through a first-order process, no matter whether $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ is present in excess or not. (3) When sufficient unreacted $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ is left at the end of the flash, a second amount of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ is decomposed at a rate 0.9 times that of the decay of Y .

The first of these experimental observations can be explained by two types of mechanisms



followed by

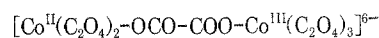


or



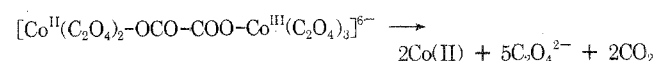
Reaction a is to some extent similar to that proposed by Parker and Hatchard for the flash photolysis of $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$.⁴ Parker did not consider however the eventual formation of a second intermediary species corresponding to X .

In the case of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ the reaction product of M with $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ could have a structure of the form



Although such a mechanism eventually could account for the data when all the $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ is immediately converted by the flash, it cannot explain the secondary conversion of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$.

The decomposition of this binuclear complex will indeed produce stable reaction products

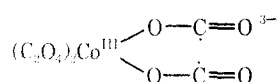


None of these reaction products can reduce another $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ molecule. Reaction a is thus in conflict with the experimental observation that Y , when decomposing, produces a species that is so reactive toward $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$.

that both the secondary conversion of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ and the decomposition of Y proceed at equal rates. Therefore reaction b is preferred. In this reaction the intermediate species Y is photochemically active.

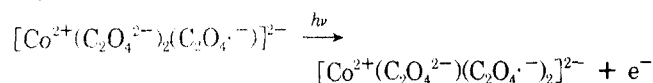
The presence of an excess $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ will inhibit the formation of X by its inner-filter effect. This scheme is in fact the same as the one withheld by Shagisultanova and Posnjak^{10,11} (see Literature Survey) on the basis of a fully different kind of work.

Our spectroscopic data give no direct indication as to the identity of Y and X. There are however a number of arguments which can be enlightening as to these identities. It is improbable that Y would have a structure such as the nonreduced species proposed by Jamieson and Perone⁷ for $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ (see also Literature Survey). In our opinion such a nonreduced species should be very short lived. Indeed, a species with biradical character such as

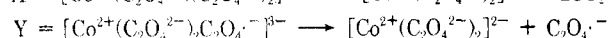
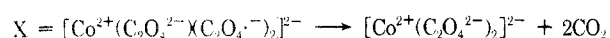


having both radical fragments linked in each others vicinity to the same metal atom, should show an outspoken tendency to recombine, thus yielding the initial reagent. Our complete immediate conversion experiments rule out such a back reaction.

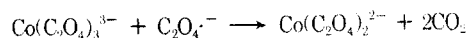
Since our experiments also rule out the possibility of a binuclear species, and in view of the well-established ligand-to-metal charge transfer nature of the actinic light, it seems logical to propose that the primary photochemical act will be the transfer of an electron from the ligand to $\text{Co}(\text{III})$ leading to an intermediary species with the structure $[\text{Co}^{2+}(\text{C}_2\text{O}_4)_2\text{C}_2\text{O}_4^{\cdot-}]^{3-}$. Such a structure has already been proposed as intermediate in the photoreduction of $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$.⁴ This species would, upon irradiation, produce species X. It is much more difficult to propose a structure for X. The work of Shagisultanova and Posnjak^{10,11} suggests that the reactive intermediate in the photolysis of $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ undergoes a second photochemical reaction, producing a solvated electron. This leads us to propose the following reaction



The two monomolecular decomposition reactions of these intermediates will then be



When no excess $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ is present the oxalate radical (species Z) will disappear in a dismutation reaction with its own species. If however there is $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ left after the flash photolysis there will be a very fast reduction of a second Co^{III} .



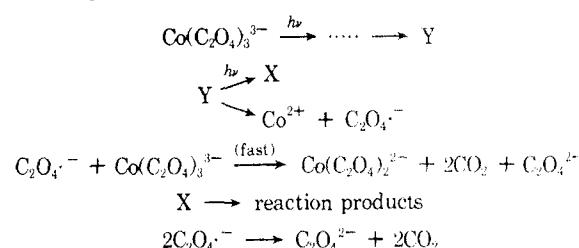
This reaction must be fast since $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ disappears with the same rate as the decomposition of Y.

Conclusions

In conditions of incomplete immediate conversion, the reaction mechanism proposed by Copestake and Uri⁸ is qualitatively in agreement with our results.

When very high light intensities are used or very low initial reagent concentrations, another intermediary species is observed. Its formation can only be explained by accepting that it results from the photolysis of the primary radical. This biphotonic process can be observed only when the $\text{Co}(\text{C}_2\text{O}_4)_3^{3-}$ concentration has become sufficiently low to avoid its inner-filter effect.

The complete reaction scheme is



There are sound arguments to accept that the structure of Y is $[\text{Co}^{2+}(\text{C}_2\text{O}_4^{2-})_2\text{C}_2\text{O}_4^{\cdot-}]^{3-}$. In line with this, the structure of X would be $[\text{Co}^{2+}(\text{C}_2\text{O}_4^{2-})(\text{C}_2\text{O}_4^{\cdot-})_2]^{2-}$.

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